Core AI Types in Cybersecurity

CompTIA SecAI+ | Lesson 1.1.1 — CertMaster Study

MODULE 1

FOUNDATIONAL CONCEPTS

What is Artificial Intelligence?

Machines capable of **problem-solving and analysis** — with minimal human direction.

Learns from Data

Identifies patterns humans might miss

Acts Autonomously

Makes decisions without step-by-step instructions

Transforms Security

Fundamentally reshaping cybersecurity practices

AI in Cybersecurity

Benefits

Threat Detection

Identify attacks faster and more accurately

Predictive Analytics

Anticipate risks before they materialize

SOC Automation

Streamline repetitive security operations

Risks

New Vulnerabilities

AI systems introduce complex attack surfaces

Governance Gaps

Inadequate oversight of AI decision-making

Flawed Algorithms

Biased or misconfigured models cause failures

Training Data Risk

Sensitive data mishandled during model training

6 Key AI Concepts for Cybersecurity

Generative AI

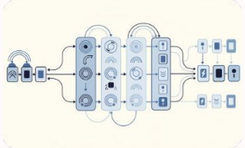
Creates new content from learned patterns — text, code, images

Machine Learning

Learns from historical data to make predictions and classifications

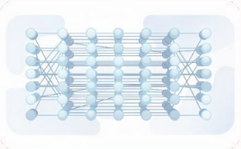
Statistical Learning

Explains *why* models make specific predictions — interpretability



Transformers

Breakthrough architecture powering modern AI sequence tasks



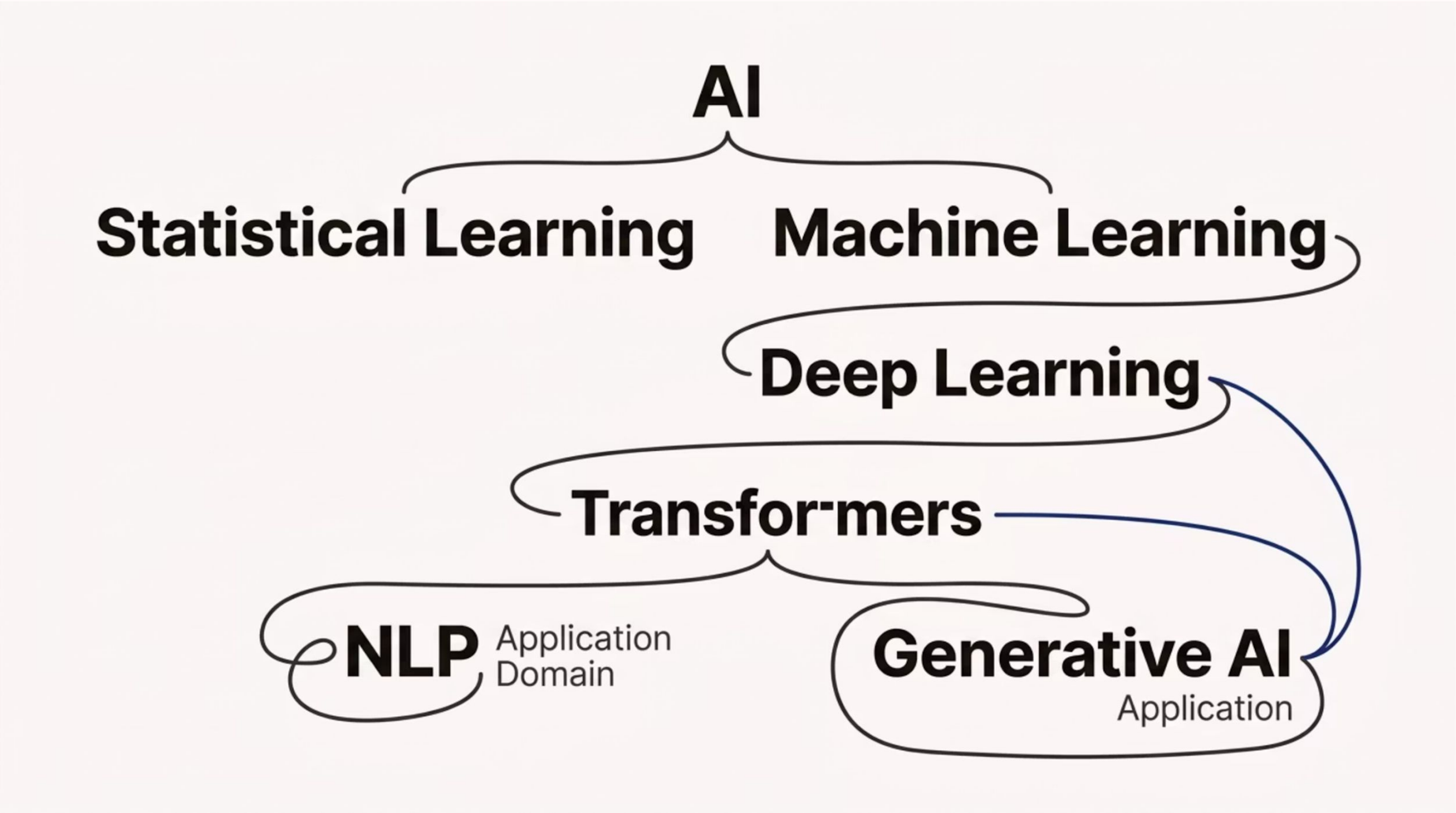
Deep Learning

Multi-layered neural networks solving highly complex tasks

NLP

Enables machines to read, interpret, and generate human language

How These AI Types Relate



AI is a broad field — each concept builds on or extends the one above it. Understanding this hierarchy helps security professionals pinpoint where vulnerabilities and capabilities emerge.

Why This Matters

> Security professionals who understand AI can harness its power — and contain its risks.

Harness the Power

Deploy AI tools effectively across the security stack

Identify the Risks

Recognize where AI introduces new attack vectors

Stay Proactive

Mitigate AI-associated risks before they are exploited